

SAMPSON COUNTY AP, NC, USAWMO: **722073**Lat: **34.976N**Lon: **78.364W**Elev: **45**StdP: **100.78**Time Zone: **-5.00 (NAE)**Period: **04-19**WBAN: **03727****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.2	-2.9	-15.0	1.0	-2.0	-12.3	1.3	1.3	8.5	11.2	7.5	10.2	1.4	10	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	35.1	25.0	33.8	24.6	32.6	24.2	26.7	31.8	26.1	31.1	25.6	30.2	2.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	20.8	28.3	24.9	20.1	27.9	24.4	19.4	27.5	83.4	31.8	81.1	31.2	78.9	30.3	34.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	5.6	4.8	-8.6	37.3	2.3	1.5	-10.2	38.4	-11.6	39.3	-12.9	40.1	-14.5	41.2		
(5)			-9.5	28.0	2.2	2.0	-11.1	29.4	-12.4	30.6	-13.7	31.7	-15.3	33.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	7.0	8.6	11.8	17.1	21.4	25.5	27.2	26.5	23.6	17.8	11.6	9.7
	DBStd	8.28	5.93	5.39	5.28	4.35	3.52	2.57	1.86	2.09	2.91	4.64	4.71	5.07
	HDD10.0	368	131	84	44	3	0	0	0	0	0	2	35	69
	HDD18.3	1470	352	276	212	73	14	0	0	0	2	67	206	270
	CDD10.0	3058	38	44	99	216	354	466	534	511	407	245	84	61
	CDD18.3	1118	1	3	8	36	110	216	276	252	159	51	4	3
	CDH23.3	10036	2	18	82	346	953	2024	2719	2277	1214	363	28	9
	CDH26.7	4094	0	1	9	82	328	908	1259	976	448	83	1	0
Wind	WSAvg	1.6	1.8	2.0	2.1	2.1	1.6	1.4	1.2	1.0	1.4	1.4	1.6	1.5
Precipitation	PrecAvg	1381	102	83	103	89	100	130	154	172	164	92	86	81
	PrecMax	1735	181	222	168	164	162	365	322	411	541	186	223	172
	PrecMin	911	23	33	25	15	41	35	68	66	7	6	12	24
	PrecStd	206	45	45	42	41	36	69	65	84	120	59	49	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	26.0	27.9	31.0	34.7	36.7	37.2	36.3	33.9	31.5	26.1	24.9
		MCWB	18.4	18.7	17.3	19.5	23.9	23.8	25.0	25.1	23.2	23.8	18.6	20.6
	2%	DB	21.3	22.8	25.9	28.9	32.0	34.5	35.1	34.1	32.4	28.8	23.9	22.5
		MCWB	17.1	17.1	17.8	19.5	21.9	24.4	25.6	25.2	23.3	21.9	18.1	19.0
	5%	DB	18.9	20.3	23.3	27.4	30.2	32.9	33.8	32.7	31.2	27.2	22.1	20.5
		MCWB	15.9	15.4	16.2	18.1	21.7	23.8	25.1	24.8	23.3	21.0	17.6	18.2
	10%	DB	16.9	17.8	21.1	25.2	28.3	31.8	32.5	31.7	29.4	25.2	20.0	18.5
		MCWB	13.8	13.8	15.1	18.1	21.1	23.5	24.7	24.5	22.8	20.1	17.1	16.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.9	20.2	20.1	23.4	25.1	26.7	27.4	27.4	26.2	25.3	21.5	22.0
		MCDB	21.5	23.2	24.4	26.4	28.3	32.4	32.9	31.6	30.6	28.3	23.2	23.5
	2%	WB	18.3	18.8	19.0	21.8	23.8	25.9	26.7	26.6	25.3	24.2	20.1	20.1
		MCDB	20.2	21.7	23.1	25.2	28.6	31.2	32.1	31.3	29.4	26.7	21.9	21.2
	5%	WB	16.3	16.9	17.8	20.5	23.0	25.3	26.2	26.0	24.6	23.2	18.8	18.4
		MCDB	18.6	19.2	21.3	24.1	27.5	30.3	31.4	30.5	28.3	25.5	20.9	19.9
	10%	WB	14.3	14.6	16.6	19.3	22.2	24.7	25.7	25.4	23.8	21.9	17.2	16.6
		MCDB	16.5	17.5	20.3	23.2	26.4	29.1	30.5	29.5	27.1	24.0	19.4	18.3
Mean Daily Temperature Range	5% DB	MDBR	10.7	11.5	12.2	12.7	11.8	11.0	10.4	9.8	10.1	11.4	11.9	10.6
		MCDBR	13.0	13.9	14.9	15.0	13.7	12.7	11.8	11.1	11.9	12.1	13.0	11.4
		MCWBR	9.3	8.9	7.6	6.6	5.2	4.1	3.7	3.6	4.3	5.7	7.7	8.1
	5% WB	MCDBR	11.3	12.5	12.2	11.5	11.1	10.9	10.5	9.9	9.7	8.9	10.4	10.3
		MCWBR	9.5	9.3	7.7	6.2	4.8	4.2	4.0	3.7	4.1	4.8	7.9	8.5
Clear-Sky Solar Irradiance	taub		0.320	0.333	0.366	0.404	0.462	0.500	0.540	0.520	0.428	0.382	0.341	0.320
	taud		2.486	2.443	2.382	2.274	2.153	2.092	1.997	2.047	2.301	2.401	2.473	2.521
	Ebn at Noon		885	911	905	882	831	797	761	769	831	843	852	863
	Edh at Noon		85	98	114	133	152	162	177	165	122	102	85	78
All-Sky Solar Radiation	RadAvg		2.66	3.34	4.41	5.53	6.06	6.37	6.07	5.45	4.60	3.80	2.93	2.34
	RadStd		0.17	0.35	0.32	0.36	0.46	0.40	0.37	0.30	0.36	0.46	0.23	0.19